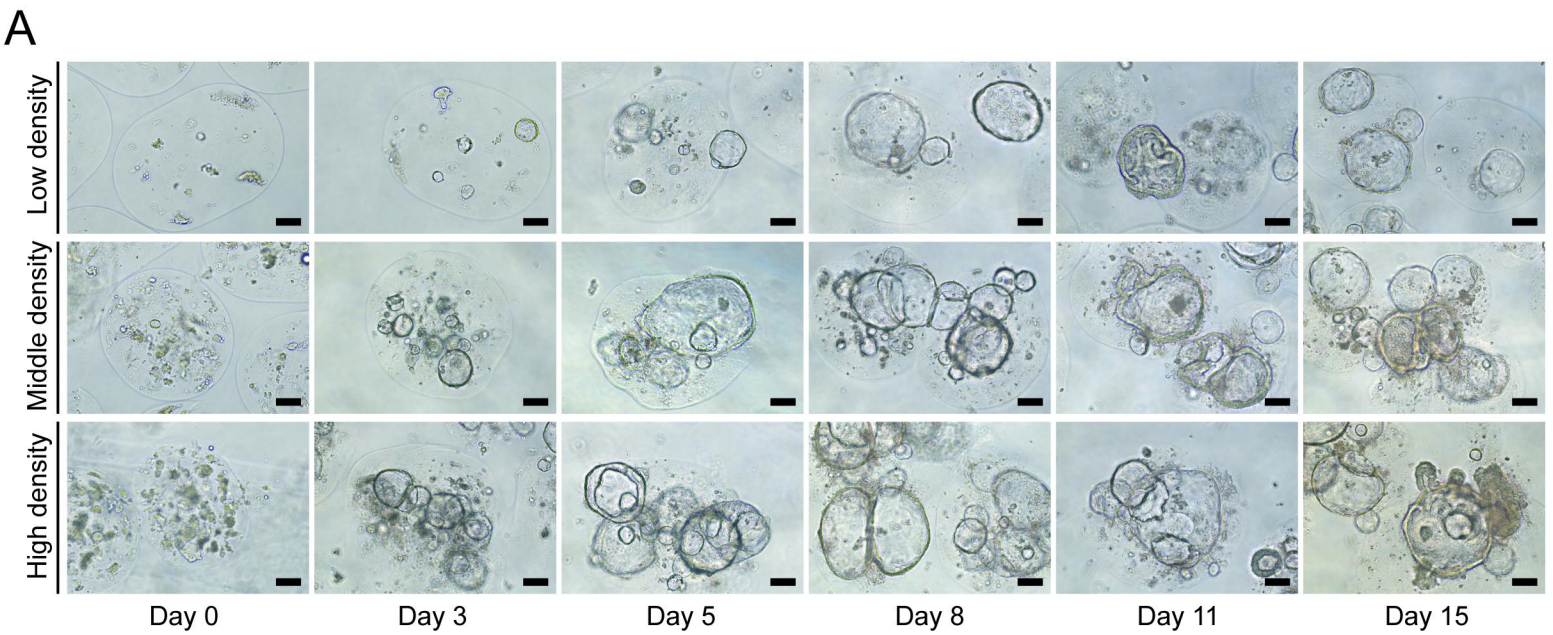
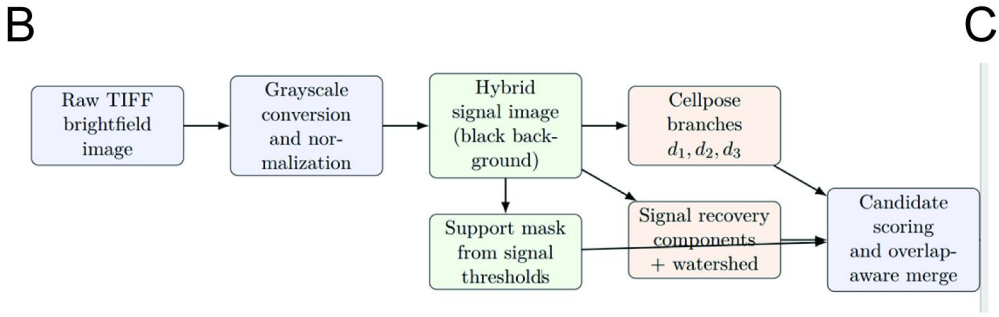
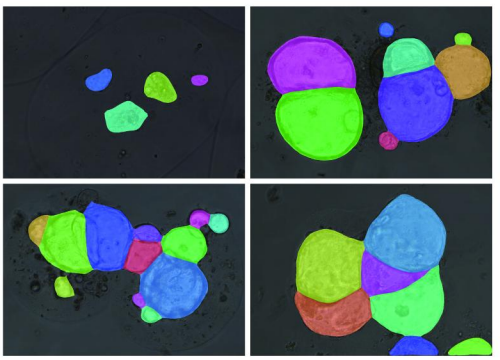
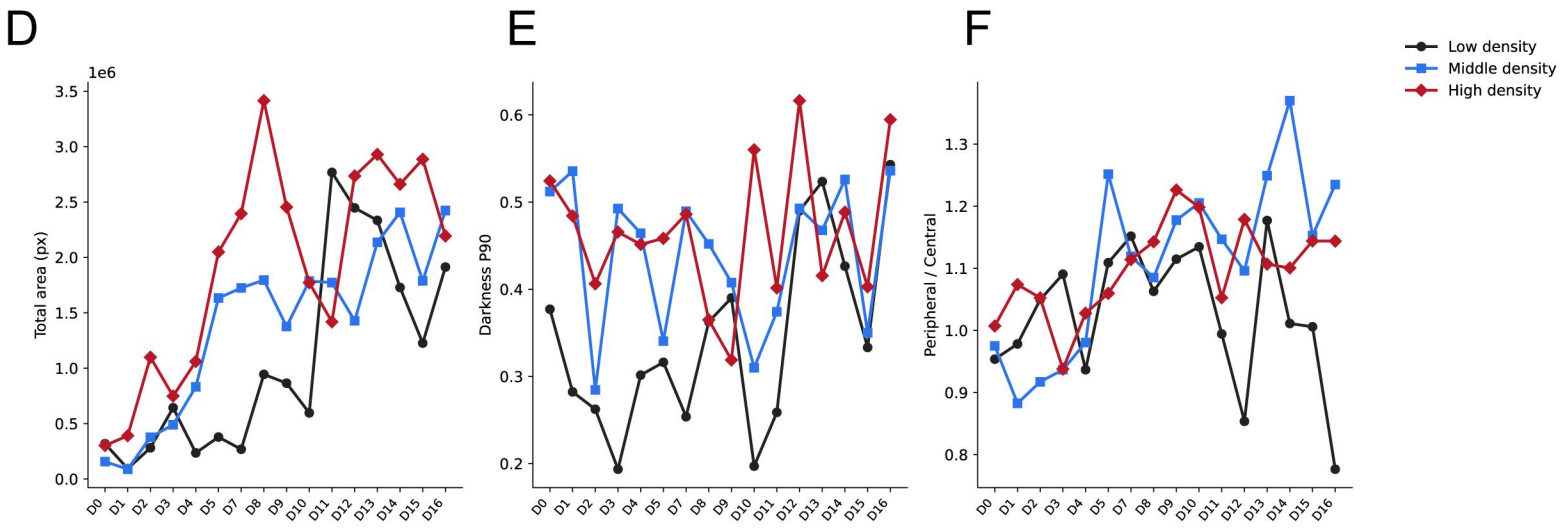

A
Low density
Middle density
High density
Day 0
Day 3
Day 5
Day 8
Day 11
Day 15


B
Raw TIFF brightfield image
Grayscale conversion and normalization
Hybrid signal image (black background)
Cellpose branches d_1, d_2, d_3
Support mask from signal thresholds
Signal recovery components + watershed
Candidate scoring and overlap-aware merge


C


D
E
F
Low density
Middle density
High density
1e6
Total area (px)
Darkness P90
Peripheral / Central
D0 D1 D2 D3 D4 D5 D7 D8 D9 D10 D11 D12 D13 D14 D15 D16